The Extension Graph of Spoof Lehmer Factorizations Is a Forest

G. Molnar

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Abstract

Hasanalizade recently showed that, given a spoof factorization F satisfying $k\tilde{\varphi}^*(F) = \tilde{\epsilon}(F) + \alpha$ for $\alpha = \pm 1$, the search for two-factor extensions $F \cdot x \cdot y$ satisfying the k-Lehmer condition $k\tilde{\varphi}^*(F \cdot x \cdot y) = \tilde{\epsilon}(F \cdot x \cdot y) - 1$ reduces to a single integer factorization. We study the global structure that this extension operation imposes on the set of k-Lehmer factorizations. We prove three results: (i) positive-integer extensions arise exclusively from $\alpha = +1$ seeds, (ii) the descent formula relating the evaluation of a Lehmer factorization to any descended pair is *independent of* k, and (iii) the resulting extension graph $\mathcal{G}_k$ is a **forest** for *every* $k \geq 2$—that is, no k-Lehmer factorization with factors ≥ 2 admits more than one descended pair. A computational census for $k = 2$ classifies all 39 factorizations reachable through seeds with at most six factors, identifies exactly 9 primitive factorizations, and reveals 17 plus-seeds, of which only six belong to the Fermat partial-product chain.

1 Introduction

Molnar and Singh [**?**] introduced the study of *spoof Lehmer factorizations*: multisets $F = \{x_1, \ldots, x_r\}$ of positive integers satisfying

$$k\,\tilde{\varphi}^*(F) = \tilde{\epsilon}(F) - 1, \qquad k \in \mathbb{Z}_{>0}, \tag{1}$$

where $\tilde{\epsilon}(F) = \prod x_i$ and $\tilde{\varphi}^*(F) = \prod(x_i - 1)$. Hasanalizade [**?**] proved that if a factorization F satisfies the auxiliary condition $k\,\tilde{\varphi}^*(F) = \tilde{\epsilon}(F) + \alpha$ for $\alpha \in \{-1, +1\}$, then the search for two-factor extensions $F \cdot x \cdot y$ satisfying (**??**) reduces to factoring a single explicit integer (Equation (**??**) below).

This note investigates the global structure that the Hasanalizade extension imposes on the solution set. The main contributions are:

(i) A proof that only $\alpha = +1$ seeds can produce extensions with positive factors (Theorem **??**).

(ii) A closed-form descent criterion (Theorem **??**) and the observation that it is *independent of* k (Corollary **??**).

(iii) A proof that the extension graph $\mathcal{G}_k$ is a forest for every $k \geq 2$ (Theorem **??**).

(iv) A computational census for $k = 2$ classifying primitives and plus-seeds (Theorem **??**).

2 Preliminaries

We follow the notation of [**?**] and [**?**]. A *spoof factorization* is a finite multiset $F = x_1 \cdots x_r$ of positive integers with $x_i \geq 2$, with *spoof evaluation* $\tilde{\epsilon}(F) = \prod x_i$ and *spoof unitary totient* $\tilde{\varphi}^*(F) = \prod(x_i - 1)$. We write $|F| = r$ for the number of factors. A factorization F with $|F| \geq 2$ is *unitary k-Lehmer* if (**??**) holds.

Definition 1. A factorization F is an *α-seed* (with respect to k) if $k\,\tilde{\varphi}^*(F) = \tilde{\epsilon}(F) + \alpha$, where $\alpha \in \{-1, +1\}$. When $\alpha = +1$, we call F a *plus-seed*.

Hasanalizade's Theorem 3 in [**?**] states that if F is an α-seed with $N = \tilde{\epsilon}(F)$, then $F \cdot x \cdot y$ is k-Lehmer if and only if

$$(\alpha x - N - \alpha)(\alpha y - N - \alpha) = N^2 + \alpha N - \alpha. \tag{2}$$

We also use the *deficiency function* $g(F) = k\,\tilde{\varphi}^*(F) - \tilde{\epsilon}(F)$, which satisfies the recurrence

$$g(F \cdot a) = (a - 1)\,g(F) - \tilde{\epsilon}(F), \qquad g(\emptyset) = k - 1. \tag{3}$$

Lehmer factorizations correspond to $g = -1$, plus-seeds to $g = 1$.

3 Only plus-seeds produce positive extensions

Theorem 2. *Let $k \geq 2$ and let F be a (-1)-seed with $\tilde{\epsilon}(F) = N \geq 1$. Then Equation (**??**) has no solutions with $x, y \geq 2$.*

Proof. For $\alpha = -1$, Equation (**??**) becomes $(x + N - 1)(y + N - 1) = N^2 - N + 1$. If $x, y \geq 2$ then the left side is at least $(N + 1)^2 = N^2 + 2N + 1 > N^2 - N + 1$ for all $N \geq 1$. □

4 The descent formula is k-independent

Definition 3. Let G be a k-Lehmer factorization. An unordered pair $\{a, b\}$ of factors of G (counted with multiplicity) is a *descended pair* if $F = G \setminus \{a, b\}$ is a plus-seed from which G arises via Equation (**??**). We call G *primitive* if it admits no descended pair, and *derived* otherwise.

Theorem 4 (Descent formula)**.** *Let $k \geq 2$ and let G be a k-Lehmer factorization with $\tilde{\epsilon}(G) = N$. An unordered pair $\{a, b\}$ of factors of G is a descended pair if and only if*

$$N(a + b - 1) = ab\big((a - 1)(b - 1) + 1\big). \tag{4}$$

Proof. The sub-factorization $F = G \setminus \{a, b\}$ has $\tilde{\epsilon}(F) = N/(ab)$ and $\tilde{\varphi}^*(F) = \tilde{\varphi}^*(G)/\big((a-1)(b-1)\big)$. Since G is k-Lehmer and F is a plus-seed:

$$k\,\tilde{\varphi}^*(F) = \frac{k\,\tilde{\varphi}^*(G)}{(a - 1)(b - 1)} = \frac{N - 1}{(a - 1)(b - 1)}, \qquad k\,\tilde{\varphi}^*(F) = \frac{N}{ab} + 1.$$

Equating:

$$\frac{N - 1}{(a - 1)(b - 1)} = \frac{N}{ab} + 1. \tag{5}$$

Clearing denominators: $(N-1)\,ab = N(a-1)(b-1) + ab(a-1)(b-1)$. Expanding $N(a-1)(b-1) = N(ab - a - b + 1)$ and collecting:

$$N(a + b - 1) = ab\big(ab - a - b + 2\big) = ab\big((a - 1)(b - 1) + 1\big).$$ □

Corollary 5 (k-independence)**.** *The descent formula (**??**) is independent of k. Equivalently, whether a pair $\{a, b\}$ can be a descended pair is determined entirely by the multiset of factors and the evaluation N, not by the value of k.*

Proof. Equation (**??**) shows that k cancels in the passage from the k-Lehmer and plus-seed conditions to (**??**). □

Remark 6. The k-independence is perhaps surprising: the Lehmer condition $k\tilde{\varphi}^* = \tilde{\epsilon} - 1$ and the plus-seed condition $k\tilde{\varphi}^* = \tilde{\epsilon} + 1$ both involve k explicitly, but dividing the first by the second at the level of a descended pair absorbs k into the relationship between $\{a, b\}$ and the remaining factors.

5 The extension graph is a forest

Definition 7. The *extension graph* $\mathcal{G}_k$ is a directed graph whose vertices are the k-Lehmer factorizations with factors ≥ 2. There is an edge $G \to F$ whenever $F = G \setminus \{a, b\}$ for some descended pair $\{a, b\}$.

Theorem 8 (Forest theorem)**.** *For every $k \geq 2$, the extension graph $\mathcal{G}_k$ is a forest: every k-Lehmer factorization with factors ≥ 2 admits at most one descended pair.*

By Corollary **??**, the descent formula (**??**) is k-independent, so the entire proof reduces to showing that no single multiset G with $\tilde{\epsilon}(G) = N$ can have two distinct unordered pairs satisfying (**??**) with an appropriate divisibility and positivity structure. The proof uses three lemmas, covering the three possible cases for the intersection of two descended pairs.

Lemma 9 (Disjoint case)**.** *Let G be a k-Lehmer factorization. Then G cannot admit two descended pairs $\{a, b\}$ and $\{c, d\}$ with $\{a, b\} \cap \{c, d\} = \emptyset$.*

Proof. Write $G = \{a, b, c, d\} \cup E$ with $\tilde{\epsilon}(E) = Q$ (where $Q = 1$ if $E = \emptyset$), so $N = abcdQ$. Applying the descent formula (**??**) to each pair and dividing by ab and cd respectively:

$$cdQ\,(a + b - 1) = (a-1)(b-1) + 1, \tag{6}$$

$$abQ\,(c + d - 1) = (c-1)(d-1) + 1. \tag{7}$$

Since $(a-1)(b-1)+1 \leq ab$ for $a, b \geq 2$ (with strict inequality unless one of a, b equals 1), multiplying gives

$$a^2b^2c^2d^2\,Q^2\,(a+b-1)(c+d-1) \leq a^2b^2c^2d^2,$$

whence $Q^2(a + b - 1)(c + d - 1) \leq 1$. Since $a + b - 1 \geq 3$ and $c + d - 1 \geq 3$, we get $Q^2 \leq 1/9 < 1$, so $Q < 1$.

If $E \neq \emptyset$ then $Q \geq 2$, a contradiction. If $E = \emptyset$ then $Q = 1$ and (**??**)–(**??**) become $cd(a+b-1) = (a - 1)(b - 1) + 1$ and $ab(c + d - 1) = (c - 1)(d - 1) + 1$. The first gives $cd < ab/(a + b - 1) \leq ab/3$; the second gives $ab < cd/(c + d - 1) \leq cd/3$. Together: $cd < ab/3 < cd/9$, which is absurd. □

Lemma 10 (Overlapping case)**.** *Let G be a k-Lehmer factorization. Then G cannot admit two descended pairs $\{a, b\}$ and $\{b, d\}$ sharing one element, where $a \neq d$.*

Proof. Write $G = \{a, b, d\} \cup E$ with $\tilde{\epsilon}(E) = Q$, so $N = abdQ$. The descent formula for each pair gives

$$dQ\,(a + b - 1) = (a-1)(b-1) + 1, \tag{8}$$

$$aQ\,(b + d - 1) = (b-1)(d-1) + 1. \tag{9}$$

Bounding: $dQ \le ab/(a+b-1)$ and $aQ \le bd/(b+d-1)$. Multiplying:

$$adQ^2 \le \frac{ab^2d}{(a+b-1)(b+d-1)}, \qquad \text{so} \qquad Q^2 \le \frac{b^2}{(a+b-1)(b+d-1)}.$$

We claim $(a+b-1)(b+d-1) > b^2$ for all $a, b, d \ge 2$. Expanding:

$$\begin{aligned}(a+b-1)(b+d-1) - b^2 &= a(b+d-1) + b(d-2) + (1-d) \\ &\ge 2(b+d-1) + 0 + (1-d) = 2b+d-1 \ge 3.\end{aligned}$$

Therefore $Q^2 < 1$, which is impossible if $E \ne \emptyset$ (since then $Q \ge 2$).

If $E = \emptyset$, then $|G| = 3$, $Q = 1$, and $G = \{a, b, d\}$. The seed of $\{a, b\}$ is the single-element factorization (d), and the plus-seed condition $k(d-1) = d+1$ gives $d = (k+1)/(k-1)$. By symmetry, the seed of $\{b, d\}$ forces $a = (k+1)/(k-1)$. We consider three sub-cases:

- $k = 2$: $d = a = 3$, $G = \{3, b, 3\}$, $N = 9b$. The Lehmer condition $2 \cdot 2(b-1) \cdot 2 = 9b - 1$ gives $8b - 8 = 9b - 1$, hence $b = -7$.
- $k = 3$: $d = a = 2$, $G = \{2, b, 2\}$, $N = 4b$. The Lehmer condition $3 \cdot 1 \cdot (b-1) \cdot 1 = 4b - 1$ gives $3b - 3 = 4b - 1$, hence $b = -2$.
- $k \ge 4$: $(k+1)/(k-1)$ is not an integer (since $k - 1 \ge 3$ and $\gcd(k-1, k+1) \mid 2$), so no single-element plus-seed exists and the $|G| = 3$ case is vacuous.

In all cases, no valid factorization arises. □

Lemma 11 (Repeated-element case)**.** *Let G be a k-Lehmer factorization. Then G cannot admit descended pairs $\{v, v\}$ and $\{v, w\}$ with $w \ne v$, where v appears with multiplicity at least two in G.*

Proof. Write $G = \{v, v, w\} \cup E$ with $\tilde{\epsilon}(E) = Q$, so $N = v^2wQ$. The descent formula for the pair $\{v, v\}$ gives, after dividing by v^2:

$$wQ(2v-1) = (v-1)^2 + 1. \tag{10}$$

The left side is a positive multiple of $2v - 1$, so $(2v-1) \mid \big((v-1)^2 + 1\big)$. We compute modulo $2v - 1$: since $2(v-1) \equiv -1 \pmod{2v-1}$,

$$4\big((v-1)^2 + 1\big) = (2v-2)^2 + 4 \equiv 1 + 4 = 5 \pmod{2v-1}.$$

As $\gcd(4, 2v-1) = 1$, we conclude $(2v-1) \mid 5$, giving $2v - 1 \in \{1, 5\}$ and $v \in \{1, 3\}$.

For $v = 3$: Equation (**??**) gives $5wQ = 5$, so $wQ = 1$, forcing $w = 1$ and $Q = 1$. But factors must be ≥ 2.

For $v = 1$: factors must be ≥ 2.

Neither case yields a valid factorization. □

Proof of Theorem **??**. Let G be a k-Lehmer factorization with factors ≥ 2, and suppose for contradiction that $\{a, b\}$ and $\{c, d\}$ are two distinct descended pairs. There are three cases:

- $\{a, b\} \cap \{c, d\} = \emptyset$: impossible by Lemma **??**.
- $|\{a, b\} \cap \{c, d\}| = 1$ with distinct values: after relabeling, $\{a, b\}$ and $\{b, d\}$ with $a \ne d$, impossible by Lemma **??**.

- $|\{a,b\} \cap \{c,d\}| = 1$ with a repeated element: $\{v,v\}$ and $\{v,w\}$, impossible by Lemma **??**.

The only remaining possibility $\{a,b\} = \{c,d\}$ contradicts the assumption that the pairs are distinct. □

Corollary 12. *For any $k \geq 2$, every derived k-Lehmer factorization is produced by a unique plus-seed.*

6 Computational census for $k = 2$

Theorem 13 (Census)**.** *For $k = 2$, the following hold among all 2-Lehmer factorizations reachable through plus-seeds with at most six factors.*

*(a) There are exactly 17 plus-seeds with $|F| \leq 6$ (including $\emptyset$). Six belong to the Fermat partial-product chain and eleven are sporadic (Table **??**).*

*(b) The resulting 39 Lehmer factorizations include exactly 9 primitives (Table **??**) and 30 derived factorizations.*

(c) The number of derived factorizations produced by a plus-seed F equals the number of divisor pairs of $\Delta_+(N) = N^2 + N - 1$, where $N = \tilde{\epsilon}(F)$.

Proof. By exhaustive enumeration using the recurrence (**??**). Starting from $g(\emptyset) = 1$, factors are added with the constraint that g remains a positive integer at every non-terminal step. The search space is bounded by Hasanalizade's Theorem 1. The forest property is confirmed *a posteriori* (and now also *a priori* by Theorem **??**). □

Table 1: Plus-seeds for $k = 2$ with $|F| \leq 6$.

$\lvert F\rvert$	Seed F	$\tilde{\epsilon}(F)$	Type
0	$\emptyset$	1	Fermat
1	(3)	3	Fermat
2	$(3,5)$	15	Fermat
3	$(3,5,17)$	255	Fermat
4	$(3,5,17,257)$	65 535	Fermat
	$(5,5,5,43)$	5 375	Sporadic
5	$(3,5,17,257,65537)$	4.29×10^9	Fermat
	$(3,5,17,353,929)$	8.36×10^7	Fermat-ext.
	$(3,9,9,23,131)$	7.32×10^5	Sporadic
	$(5,5,5,43,5377)$	2.89×10^7	Sporadic
	$(5,5,5,47,453)$	2.66×10^6	Sporadic
	$(5,7,7,7,133)$	2.28×10^5	Sporadic
6	$(3,9,9,23,131,732161)$	5.36×10^{11}	Sporadic
	$(3,11,11,11,677,3659)$	9.89×10^9	Sporadic
	$(5,5,5,53,215,158265)$	2.25×10^{11}	Sporadic
	$(5,7,7,7,133,228097)$	5.20×10^{10}	Sporadic
	$(5,7,7,13,13,619)$	2.56×10^7	Sporadic

Table 2: Primitive 2-Lehmer factorizations.

$\lvert G\rvert$	Primitive G	$\tilde{\epsilon}(G)$
5	$(5,5,5,43,5375)$	2.89×10^{7}
5	$(5,5,9,9,89)$	1.80×10^{5}
6	$(3,5,17,377,1217,2295)$	2.69×10^{11}
6	$(3,5,17,395,1059,2303)$	2.46×10^{11}
6	$(3,5,33,53,69,8343)$	1.51×10^{10}
6	$(3,9,9,23,131,732159)$	5.36×10^{11}
6	$(5,5,5,47,457,50659)$	1.36×10^{11}
6	$(5,5,5,65,129,2325)$	2.44×10^{9}
6	$(5,7,7,7,133,228095)$	5.20×10^{10}

7 The Fermat chain and sporadic seeds

The Fermat partial products $F^{(s)}=\prod_{i=0}^{s-1}(2^{2^i}+1)$ form a linear chain of plus-seeds for $k=2$, since the telescoping identity

$$2\prod_{i=0}^{s-1}2^{2^i}=2^{2^s}=\prod_{i=0}^{s-1}(2^{2^i}+1)+1$$

gives $2\,\tilde{\varphi}^*(F^{(s)})=\tilde{\epsilon}(F^{(s)})+1$. Each seed generates Lehmer factorizations by factoring $\Delta_+(2^{2^s}-1)=2^{2^{s+1}}-2^{2^s}-1$.

However, the Fermat chain accounts for only 6 of the 17 plus-seeds in our census. One seed, $(3,5,17,353,929)$, bridges the two worlds: it arises from the Fermat seed $(3,5,17)$ via a *seed-to-seed extension*, solving $(x-256)(y-256)=255^2+255-1=65279=97\cdot673$ to give $x=353$ and $y=929$. This suggests a secondary recursive structure on the set of plus-seeds themselves—a direction for future work.

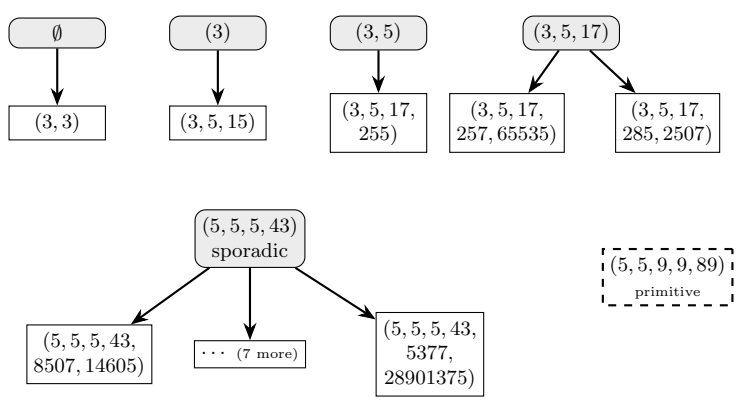

Figure 1: A fragment of $\mathcal{G}_2$. Rounded: plus-seeds. Solid: derived Lehmer factorizations. Dashed: primitive.

8 Open questions

1. **Finiteness of primitives.** All nine known primitive 2-Lehmer factorizations have $|G|\geq5$. Are there finitely many for each k?

2. **Realizability.** Which spoof factorizations correspond to actual integers $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ with each $x_i = p_i^{\alpha_i}$ a prime power?

3. **Sporadic seed structure.** Is there a generating mechanism for the sporadic plus-seeds, or a secondary recursive structure explaining their proliferation?

4. **Density of primitives.** Among our 39 factorizations, 9 are primitive ($\approx$23%). Does this proportion converge?

References

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